

CONTEXT

You are employed by a Swiss small/medium business making heavy use of data. You are member of a team of data engineers and data scientists. Your boss is intrigued by open source software and cloud computing services for data engineering and data science.

Your boss asks that you investigate, in a small group of colleagues, an open source tool or cloud service that could be of interest to your company. She wants you to prepare a presentation, for her and the product development team, that explains the service / technology. She also wants you to develop an application showing the benefits in a small example use case. You are asked to demo the example use case in the second part of your presentation.

INSTRUCTIONS

- Form a group of 4 students and declare it in Teams (Team *IST*, channel *Labos*, file *Workshop groups*).
- Select your preferred topic and a second and third choice. Declare your preferences in Teams (Team *IST*, channel *Labos*, file *Workshop topic preferences*). Put your three choices into the column corresponding to your group. Put a value of **1** into the cell corresponding to your first choice, **2** for your second choice, **3** for your third choice. When all groups have declared their choices, the professor will resolve any conflicts.
- Schedule for the remainder of the course:

Week	Cours	Labo
S14	TE2	Work on topic
S15	Work on topic	Work on topic
S16	Work on topic	Presentations

- Give a live presentation of maximum 12 minutes (theoretical part and demo of example use case). It will be followed by questions (6 minutes).
 - Deliver your slides by Friday June 12, 16:15 so that the professor can make it available to the class.
- Explain globally your topic. Recall that your audience consists of data engineers and data scientists, but also business people (your boss). Don't forget to discuss cost and strategic questions.



- Be critical! Avoid repeating marketing claims ("enjoy the best developer experience").
- When discussing cost, invent a realistic cost scenario and give the monthly cost.
- Discuss the risk of vendor lock-in.
- For the example use case choose a practical scenario relevant for a small/medium business that illustrates well the strengths and weaknesses of the cloud service/technology. (NB: This is not a mini-project, you are not asked to implement a functional application.)
- Give general recommendations when it makes sense to use the cloud service / technology and when it doesn't.
- Give pointers to references for further detailed information, explanations.
- Write a short document (minimum 1 page) in the style of a "cheat sheet" for an engineer that explains
 - what the cloud service/technology does and why it's interesting
 - how to get started
 - how to use it with the most important commands/operations
- You will have several sessions for preparation. During these session you can ask the professor to assist you in the preparation of the presentation and example use case.

DELIVERABLES (DUE FRIDAY JUNE 12, 16:15):

- slides of presentation (PDF)
- cheat sheet (max. 2 pages, PDF)

PROPOSED TOPICS

1. Amazon Redshift cloud data warehouse

- [Website](#)
- [FAQ](#)
- [Blog_posts](#)
- [SIGMOD'22_paper](#)

2. Google BigQuery cloud data warehouse

- [Website](#)
- [FAQ](#)
- [Blog_posts](#)

3. Azure Synapse Analytics cloud data warehouse

- [Website](#)



- [FAQ](#)
- [Blog_posts](#)
- 4 . Snowflake cloud data warehouse
 - [Website](#)
 - [Blog_posts](#)
- 5 . Amazon SageMaker machine learning platform
 - [Website](#)
 - [FAQ](#)
 - [Blog_posts](#)
- 6 . Google Vertex AI machine learning platform
 - [Website](#)
 - [Blog_posts](#)
- 7 . Azure Machine Learning platform
 - [Website](#)
 - [FAQ](#)
 - [Blog_posts](#)
- 8 . Apache AirFlow
 - [Website](#)
 - [Data Pipelines with Apache Airflow](#) (access O'Reilly via <http://go.oreilly.com/hes-so>)
- 9 . Data Version Control (DVC)
 - [Website](#)
 - [GitHub repo](#)
 - [Data version control with DVC. What do the authors have to say?](#)
- 10 . Data Build Tool (DBT)
 - [Website](#)
 - [GitHub repo](#)
 - [dbt Tutorial: 7 Must-Know Concepts For Data Engineers](#)

